# DEMONSTRATING FTP VS SFTP

# SECURITY

**Introduction**

Organizations transfer files daily across networks, including documents, databases, backups, and confidential records.

Two common protocols used for file transfer are:

- FTP (File Transfer Protocol)
- SFTP (Secure File Transfer Protocol)

Although both perform the same function, they differ significantly in security.

This project demonstrates how FTP and SFTP operate and compares their security using packet analysis with Wireshark.

**Objectives**

- Demonstrate FTP and SFTP Security
- Wireshark packet capture analysis for FTP and SFTP
- Comparison of FTP and SFTP security
- Discuss the risk associated with FTP
- Explain why SFTP should be used and recommended

**Tools Used:**

| Tools | Purpose |
|---|---|
| FileZilla Server | For FTP Configuration |
| Kali Linux | FTP Client Machine/SFTP Demonstration |
| OpenSSH | SFTP Service Running |
| Wireshark | FTP and SFTP Packet Capture Analysis |

File Transfer Protocol (FTP) and Secure File Transfer Protocol (SFTP) are usually used

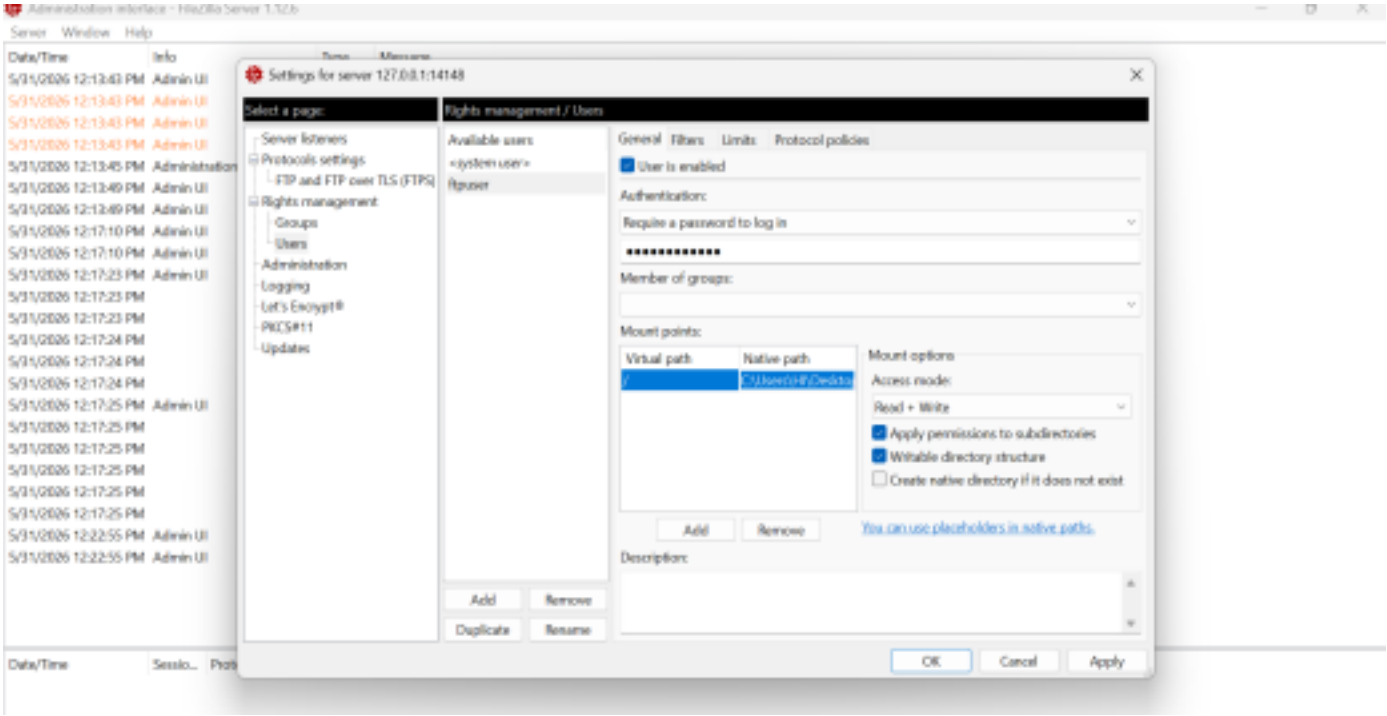
methods for transferring files across networks. FTP provides file transfer functionality but does not encrypt broadcast data, making it at risk to sniffing.

SFTP, which operates over the Secure Shell (SSH) protocol, encrypts all information transfer between the client and server, ensuring confidentiality and integrity.

This project shows the differences between FTP and SFTP security by capturing and analyzing network traffic using Wireshark. This project shows how FTP exposes confidential information while SFTP protects it through encryption.

## 1. FTP ENVIRONMENT SETUP

A local FTP environment was set up using FileZilla Server. An FTP user account (**ftpuser**) was created with a password (**ftpuser12345**). Also a shared folder was configured to allow file transfers between the FTP server and client.
A file named **myproject4.txt.txt** was set up and used during the file transfer activity.



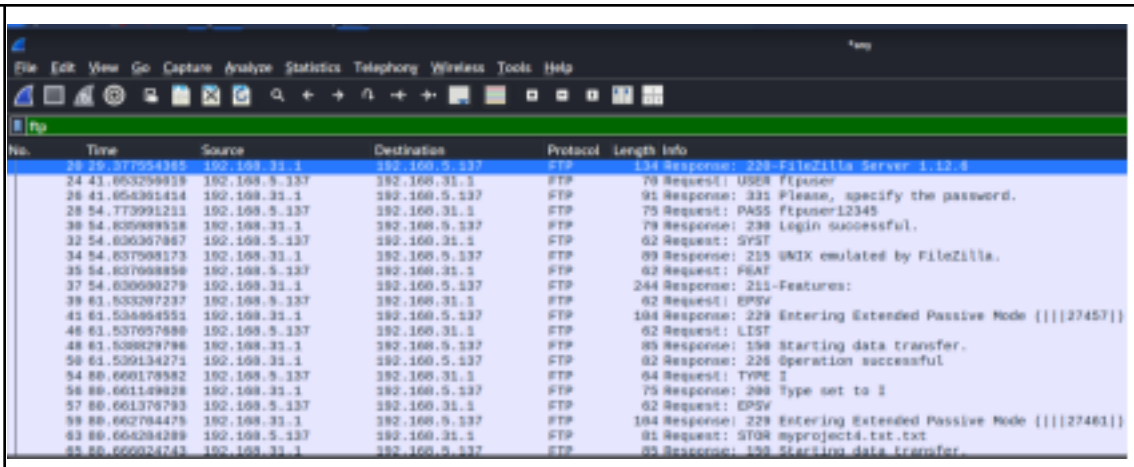

## 2. SUCCESSFUL FTP LOGIN

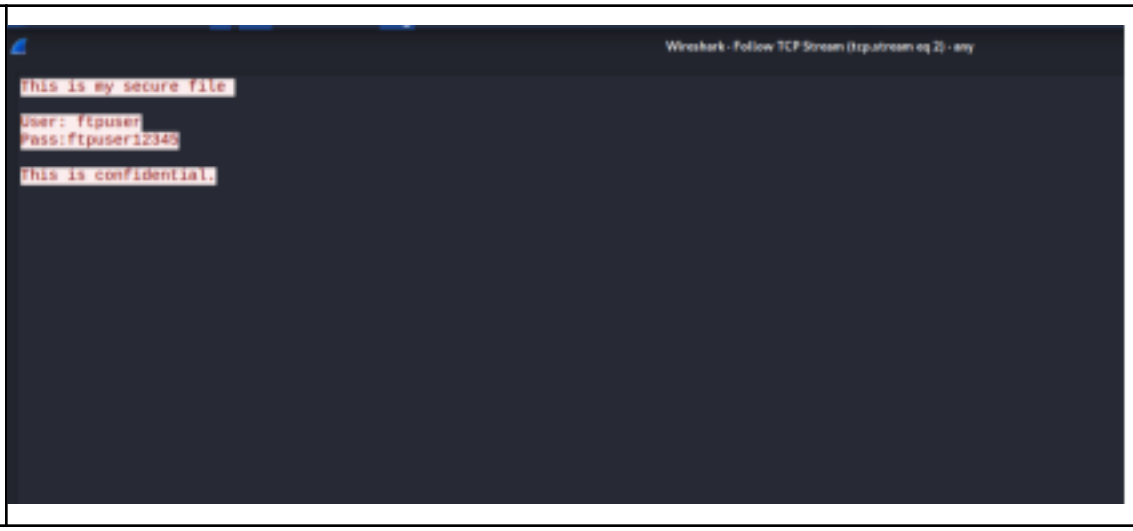
Kali Linux was used as the FTP client to link to the FileZilla Server running on Windows through the server's IP address. The Login was Successful, allowing file transfer operations.

**(myproject4.txt.txt)** was uploaded successfully through the FTP connection.

# 3. FTP TRAFFIC CAPTURE AND ANALYSIS

Wireshark was used to capture network packets generated during the FTP access period. The captured traffic was evaluated to determine whether confidential details could be viewed by an attacker tracking the network.

| | |
|---|---|
| Wireshark Showing USER Command |  |
| Wireshark Showing File Data Transfer |  |

| | |
|---|---|
| Wireshark Showing Readable File Content |  |

During the FTP interaction, Wireshark was used to record and analyze network traffic between the Kali Linux FTP client and the FileZilla FTP server. The analysis indicated that FTP transfers all data exchange in plaintext without encryption.

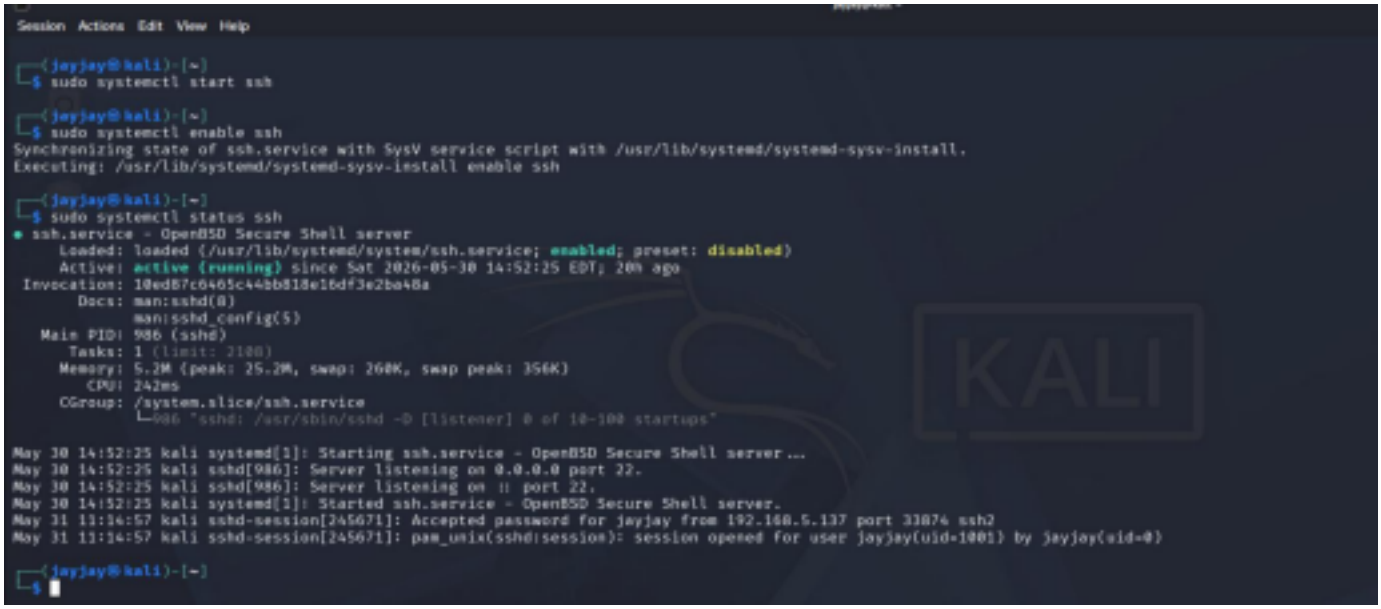
The Captured packets showed the Username and Password used for authentication through the FTP USER and PASS commands. The records of the sent file (**myproject4.txt.txt**) was visible and readable within the packet capture. This shows that any attacker monitoring the network could intercept confidential data, including login details and transferred information.

These outcomes confirm that FTP does not offer confidentiality or protection against sniffing attacks, making it an unprotected protocol for transferring sensitive information.

## SFTP ENVIRONMENT SETUP

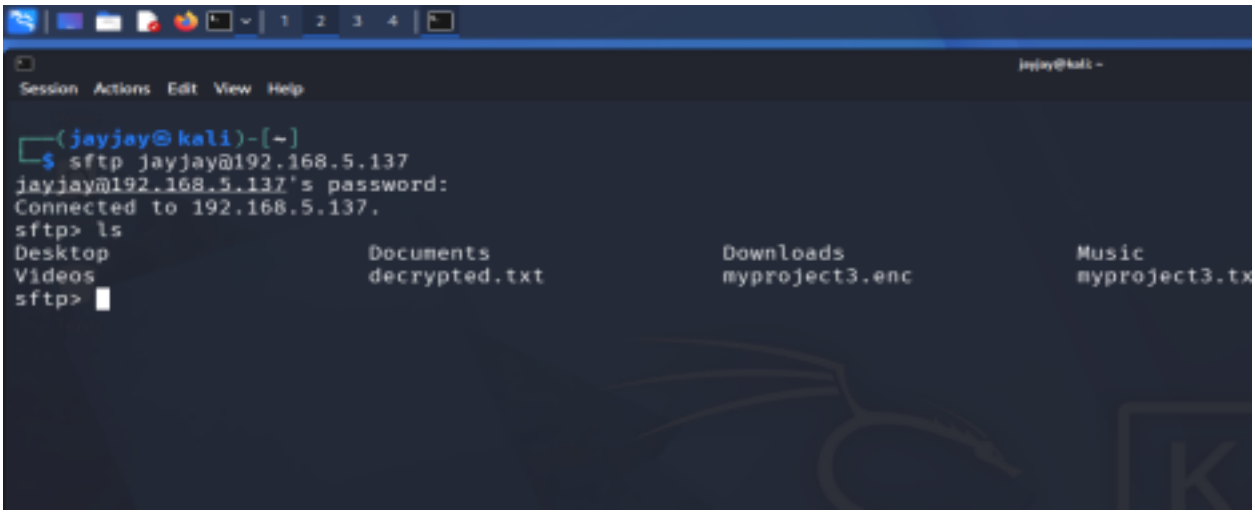
To provide secure file transfer, the OpenSSH service accessible on Kali Linux was used to set up an SFTP connection. A sample file (**sftp_groupproject4.txt**) was created specifically for the SFTP demonstration and transferred through the encrypted connection.

### 1. OpenSSH Service Running



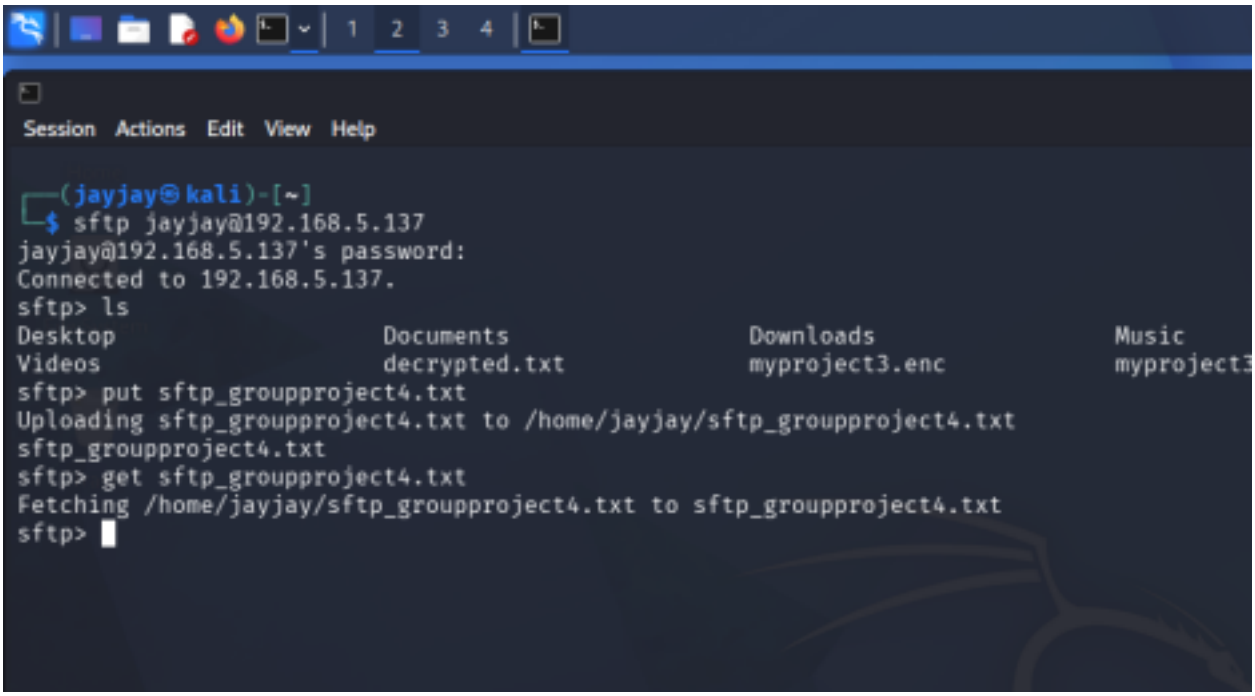

### 2. SUCCESSFUL SFTP LOGIN

The SFTP client successfully connected to the OpenSSH server running on kali Linux using the server's IP address and valid Login details.



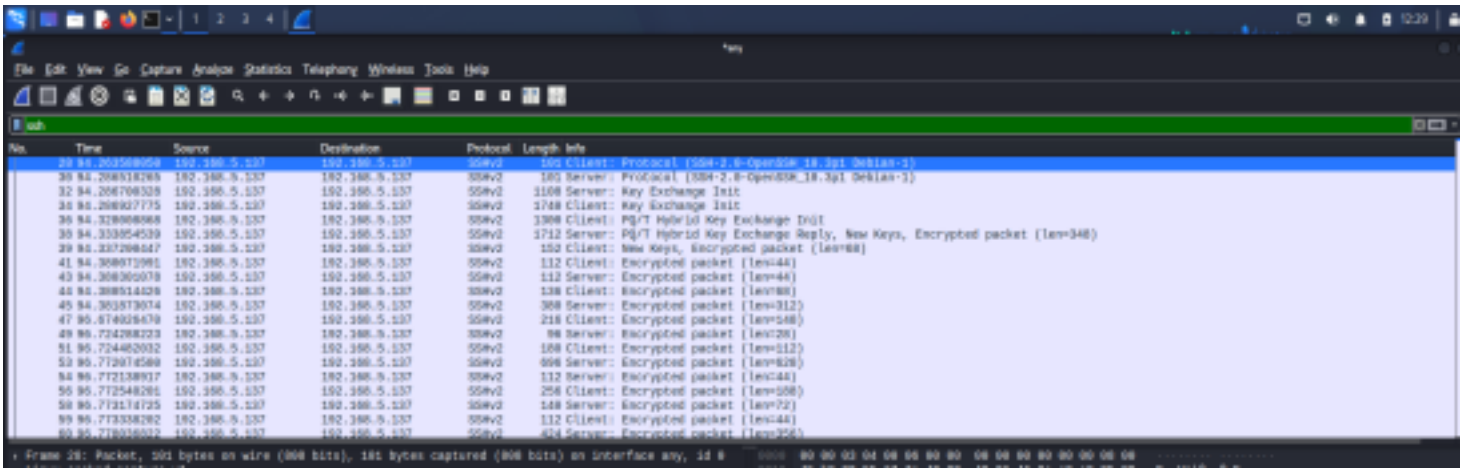

## 3. SFTP FILE TRANSFER

The SFTP text file (**sftp_groupproject4.txt**) was accurately sent through the encrypted SFTP  connection



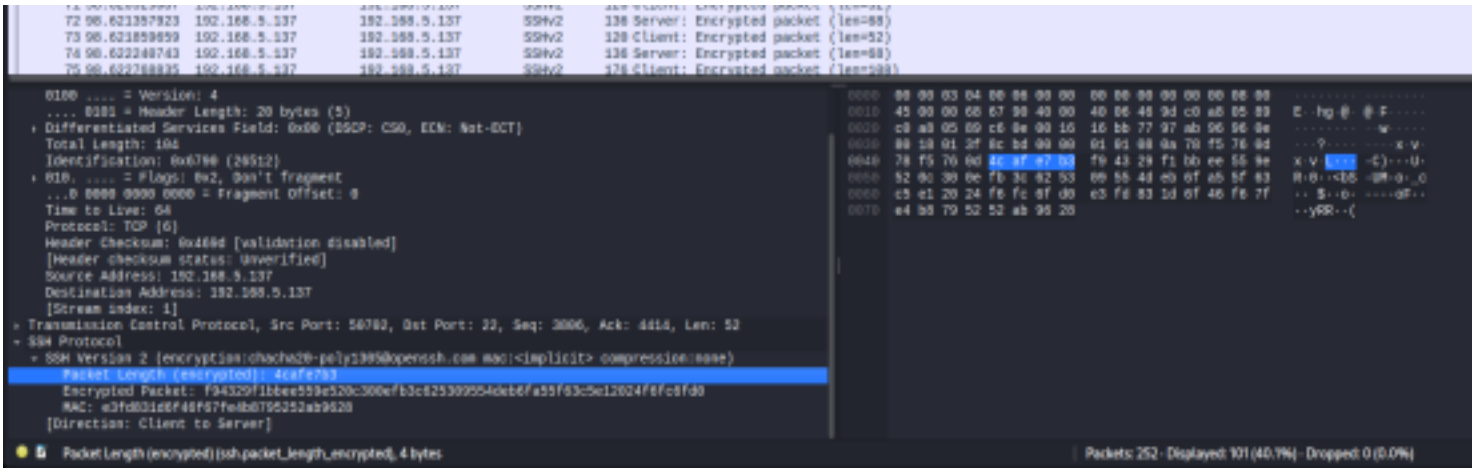

## 4. CAPTURE SSH TRAFFIC

Wireshark was used to capture network traffic generated during the SFTP connection. The packet  capture was analyzed to establish whether credentials and file contents could be viewed.



## 5. ENCRYPTED SFTP TRAFFIC

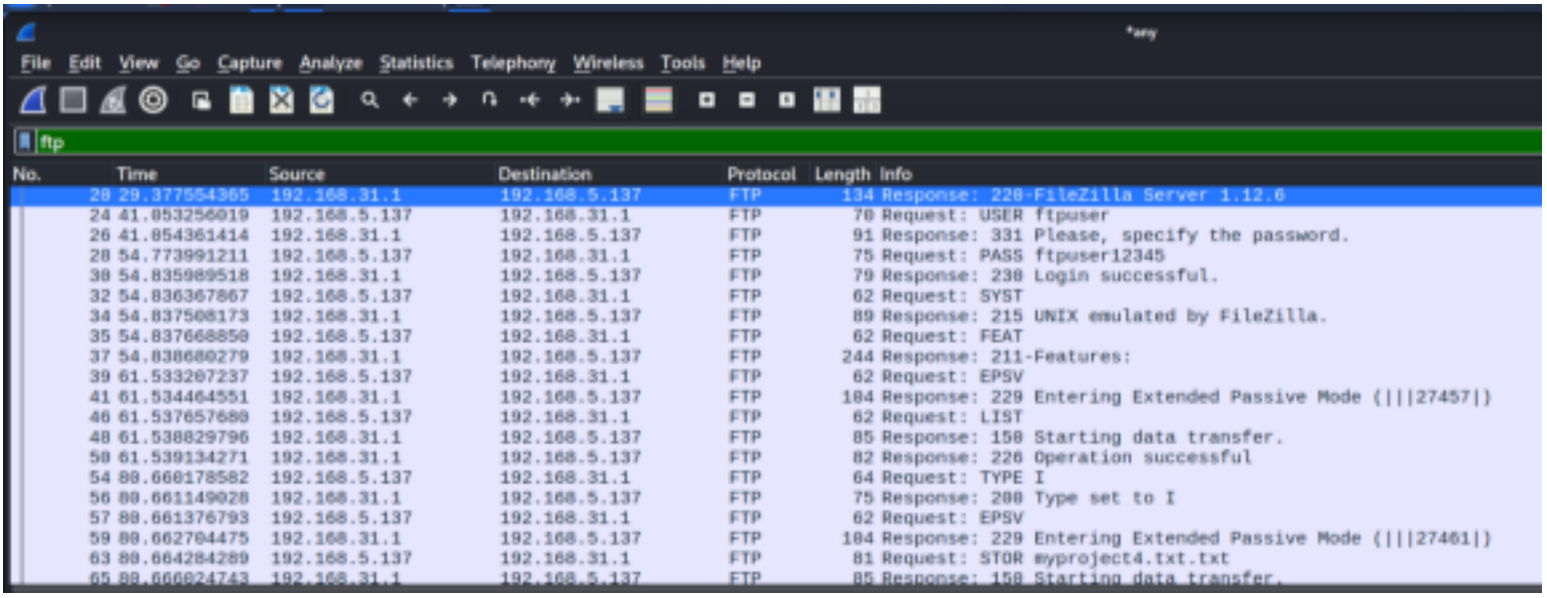
Unlike FTP traffic, the packet information appeared encrypted and unreadable. User credentials and  file contents could not be viewed within the captured packets.



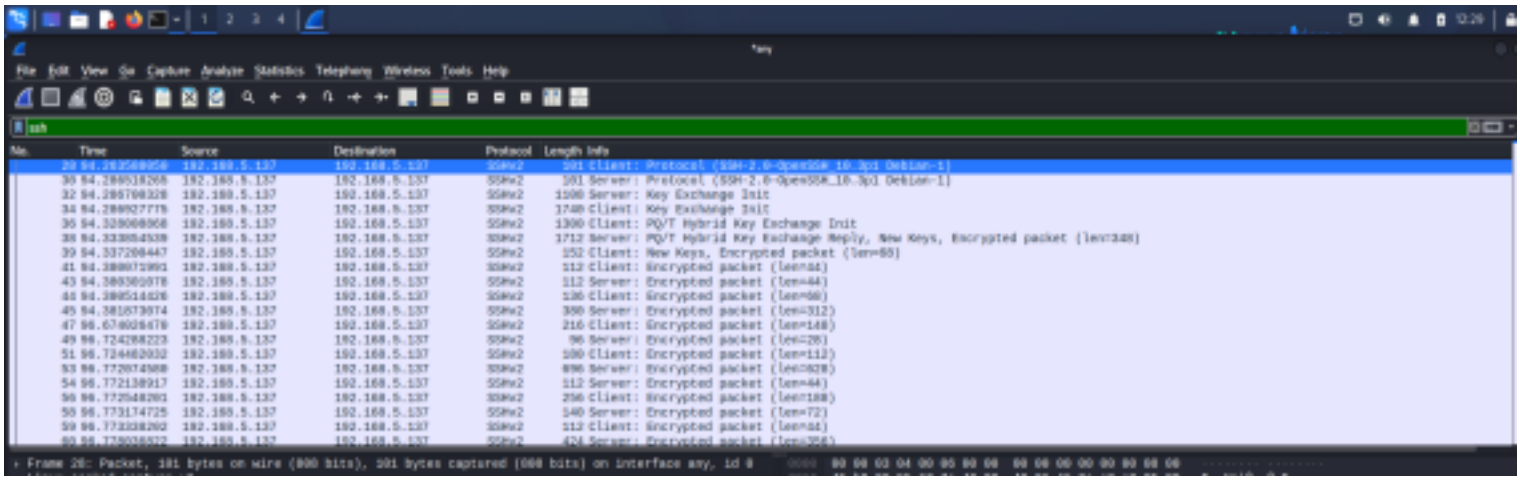
## COMPARATIVE ANALYSIS

The FTP and SFTP packet captures were compared to evaluate their security characteristics

| FTP PACKET |
| --- |




SFTP PACKET




## COMPARISON OF FTP AND SFTP TRAFFIC

| FTP | SFTP |
|---|---|
| Provides zero encryption. All commands and data pass the network interface in raw, readable plaintext. | All packets are encrypted, data and commands are all encrypted before leaving the system. |
| Wireshark FTP Capture: it showed commands and contexts of the file like USER (ftpuser) and PASS (ftpuser12345) in plaintext. | Shows the protocols SSHv2 and key exchange handshakes then all subsequent data blocks are all encrypted. |
| FTP Risks: Highly vulnerable to man-in-the middle interception(MITM), data harvesting and packet sniffing. Hackers monitoring local network interfaces can capture data and | SFTP Protection: Neutralizes sniffing threats via SSH encryption, reduces man-in-the middle attacks during sessions. It also adds integrity to ensure packets are not tampered with |

| | |
|---|---|
| credentials. | or dropped during transit. |

## SECURITY RISKS OF FTP

FTP presents several security risks because it does not offer encryption for data transferred across the network. user, pass commands, and are sent in readable data, making them detectable to attackers who record the network traffic.

On this project, Wireshark analysis displayed that FTP credentials and transmitted file records could be seen from the captured packets. This shows how attackers can disrupt confidential information through packet sniffing methods.

FTP also lacks internal methods to ensure data confidential and integrity. As a result, confidential information may be revealed, changed, or connected to by unauthorized individuals during dissemination.

These vulnerabilities make FTP unacceptable for transmitting sensitive information such as financial records, employee data, customer information, and business documents. For this reason, safeguarded choices such as SFTP are recommended because they protect data through encryption and protected verification.

The packet capture illustrates that FTP transfer authentication credentials and file contents without encryption. This makes FTP exposed to packet sniffing attacks, credential theft, and unapproved access.

## HOW SFTP PROVIDES CONFIDENTIALITY AND INTEGRITY

### Confidentiality:

Ensures that unauthorized users and attackers won't be able to access or read the data being transferred with the help of SFTP.

The encrypted SSH Protocol (SSHv2): Before data is transferred across the network, SFTP uses the SSH Protocol to build a secure route between the client and the server.

SFTP uses a two-step locking process to keep data hidden. It uses asymmetric encryption at the very beginning (the handshakes) to safely pass a temporary symmetric key to the server then switches back to symmetric encryption which is faster.

### Integrity (Anti-Tampering):

Integrity ensures that data is not altered, tampered or modified during transit.

Creating a Digital fingerprint (or Checksum): SFTP uses digital fingerprint (hashes) so hackers or unauthorized users can't alter or modify a file without getting caught. If hackers or unauthorized users tamper with the file, the fingerprint will change which tells the receiver or owner that the file has been altered. The receiver reassesses the hash to check if the hash is safe and unaltered.

## REAL-WORLD IMPLICATIONS

Banks and organizations constantly transfer payrolls files, payment instructions, operation records, and financial reports with corporate client and business partners, because these files contain confidential financial data, the banking system widely uses SFTP to protect data during communication.

Why FTP Is a Risk?

- FTP transfer usernames, passwords, and information in readable data
- Attackers observing network traffic can possibly divert logins and confidential files.
- Revealed financial data can lead to scam, regulatory penalties, and reputational damage Why SFTP Is Preferred?
- SFTP encrypts authentication credentials
- SFTP encrypts all transmitted files using SSH
- Captured network traffic is unreadable without the encryption keys
- It helps agencies meet security and fulfilment requirements.

Implications:

- Safeguarding of customer and employee monetary information
- Lessened risk of data breaches and credential fraud
- Adherence with industry security standards
- Secure communication between companies and banking partners

Enterprises regularly transmit confidential records across networks. Using FTP could leak classified information to exploiters observing network data flow. SFTP is recommended because it encrypts communications, protects user credentials, and ensures the confidentiality and integrity of transferred files.

## ENCRYPTION MECHANISM BEHIND SFTP (SSH TUNNEL)

In this project, SFTP was used to transfer files securely between the client (Kali Linux) and the SSH server. Unlike FTP, where data is seen in plaintext.

When the client (Kali Linux) initiated an SFTP connection, an SSH handshake occurred between the client and the server. During the process, encryption keys were exchanged and a secure encrypted tunnel was initiated. Once the tunnel is created, all communications between the client and the server is encrypted before being transmitted across the network in

case of man-in-the-middle sniffing.

The encryption protected usernames, passwords, file contents and commands from being intercepted by unauthorized users. As a result, when the network traffic capture tool (Wireshark) filtered the packets, the data being transferred was not in plaintext.

The packet capture showed encrypted SSH packets. This demonstrates that SFTP provides confidentiality by preventing unauthorized users from viewing sensitive information when monitoring traffic.

SFTP includes integrity verification mechanisms that ensure transferred data is not modified during transmission. Any unauthorized tampering of the data can be detected, helping to maintain the integrity of the files.

The results of the Wireshark analysis show that SFTP ensures protection of transferred data through the use of encrypted SSH tunnel, making it more secure than FTP.

## IMPORTANCE OF USING SECURE PROTOCOLS IN EVERYDAY NETWORK COMMUNICATION

- PROTECTING PRIVACY: secure protocols encrypt data in transit, guarding sensitive information like (passwords, financial details, or personal messages) from interception by unauthorized personnel. Without encryption, our usual operations like logging into our social media accounts, sending emails or even shopping online might reveal personal data to malicious actors.
- SAFEGUARDING TRANSACTIONS: online banking, e-commerce, and digital payments all use secure protocols to guarantee the integrity and privacy of financial exchanges. TLS (Transport Layer Security) helps to ensure the integrity of the file which simply means that the data reach their destination securely and intact
- PREVENTING CYBER ATTACKS: Secure protocols helps to protect against attacks like traffic interception attack where cyber criminal's intercepts and alter the communication. They also help to identify authenticity ensuring the site is genuine and not a fraudulent duplicate

**Conclusion**

This project demonstrated that:

- FTP exposes usernames, passwords, and files in plaintext.
- Wireshark can easily capture FTP credentials.
- SFTP encrypts all communications through SSH.
- SFTP provides confidentiality, integrity, and stronger security.

**Therefore, SFTP is the recommended protocol for secure file transfer in modern organizations.**